Writing, Reading and Reviewing Papers

20261103, NTNU, Trondheim

Robert Feldt



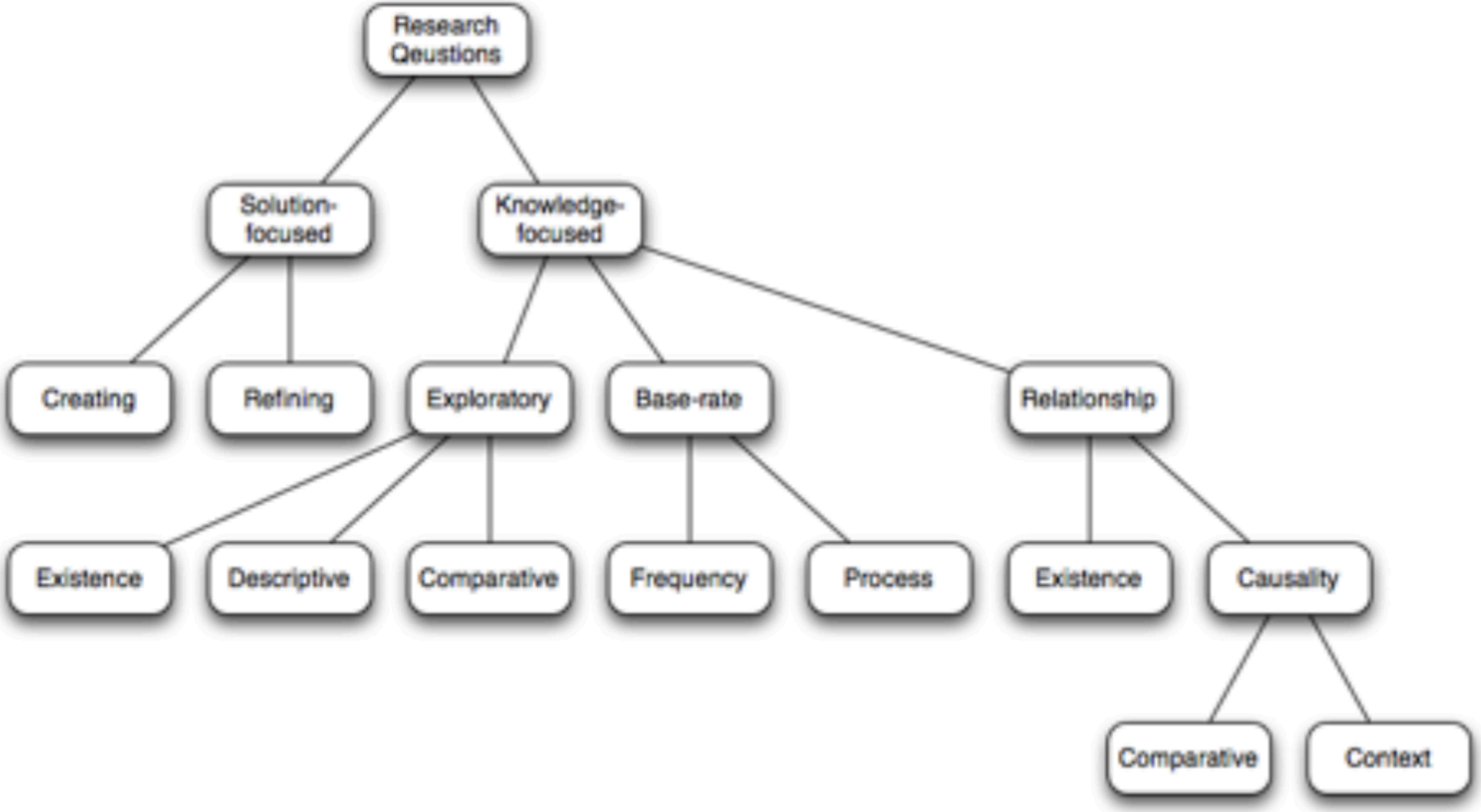
What to study in a paper?

Main Types of RQs	Answers
Exploratory	Answers give clearer understanding, better definitions of concepts, evidence that we can measure them validly
Base-rate	Answers describe the normal pattern of occurrence of the phenomena
Relationship	Answers describe if and how two phenomena are related
Solution-focused	Describes better ways to solve some problem or situation. Examples are: *“What is an effective way to achieve X?”,* *“Which strategies help achieve X?”,* *“How can we refined S to achieve X in a better way?”*

Sub-Types of RQs	Examples
Exploratory/Existence	*“Does X exist?”,* *“Is Y something that software engineers really do?”*
Exploratory/Descriptive	*“What is X like?”,* *“What are its properties/attributes?”,* *“How can we categorize/measure X?”,* *“What are the components of X?”*
Exploratory/Comparative	*“How does X differ from Y?”*
Base-rate/Frequency	“How often does X occur?”, “What is an average amount of X?”
Base-rate/Process	*“How does X normally work?”,* *“What is the process by which X happens?”,* *“In what sequence does the events of X occur?”*
Relationship/Existence	*“Are X and Y related?”,* *“Do occurrences of X correlate with Y?”* *“What correlates with X?”*
Relationship/Causality	*“What causes X?”,* *“Does X cause Y?”,* *“Does X prevent Y?”,*
Causality/Comparative	*“Does X cause more Y than Z does?”,* *“Is X better at preventing Y than Z is?”*
Causality/Context	*“Does X cause more Y under one condition than others?”*

Formulating Research Questions

“Mine” is mainly for Classifying



Alvesson is for Problematising then Formulating

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GENERATING RESEARCH QUESTIONS THROUGH PROBLEMATIZATION

MATS ALVESSON
University of Lund and University of Queensland

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University of Queensland

It is increasingly recognized that what makes a theory interesting and influential is that it challenges our assumptions in some significant way. However, established ways for arriving at research questions mean spotting or constructing gaps in existing theories rather than challenging their assumptions. We propose problematization as a methodology for identifying and challenging assumptions underlying existing literature and, based on that, formulating research questions that are likely to lead to more influential theories.

Covvey is for Validating/Checking

Back to the basics: guidance for formulating good research questions

Jordan R Covvey[a], **Chyna McClendon**[a], **Michael R Gionfriddo**[a,*]

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Abstract

Good science is driven by rigorous questions. Much like the foundation of a house, a research question must be carefully constructed to prevent downstream problems in project execution. And yet, pharmacy researchers and scholars across all career stages may find themselves struggling. The purpose of this commentary is to provide useful guidance on composing and evaluating rigorous research questions. A variety of frameworks, such as PICO (Patient/population; Intervention; Comparison; Outcome), are available to researchers and can assist them in ensuring that their research question has covered all relevant components. Additionally, the FINER (Feasible; Interesting; Novel; Ethical; and Relevant) criteria can help researchers with evaluating their research questions for practical considerations. Finally, building awareness of common pitfalls when composing research questions can aid researchers to avoid issues that they may not otherwise discover until their manuscript undergoes peer review.

Good research questions do not only fill gaps — they challenge assumptions

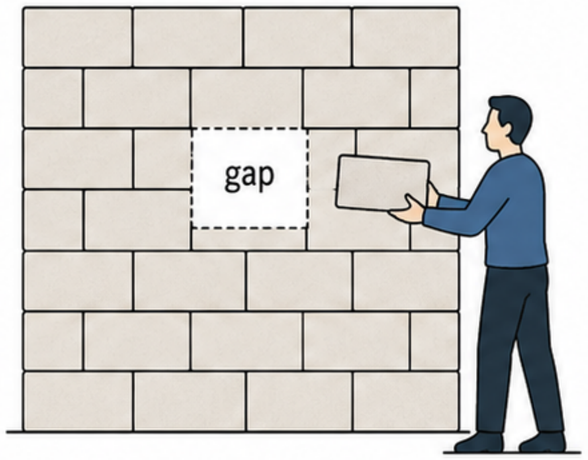
From gap-spotting to problematization

Incremental Extension

Gap-spotting

Fill the gap

Existing literature

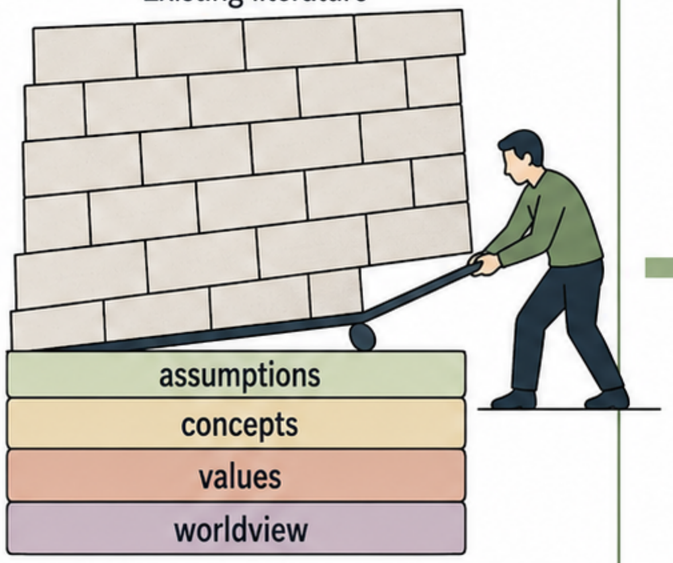


Add to the existing structure

Problematization

Shift the ground

Existing literature



Question the structure that makes the gap appear

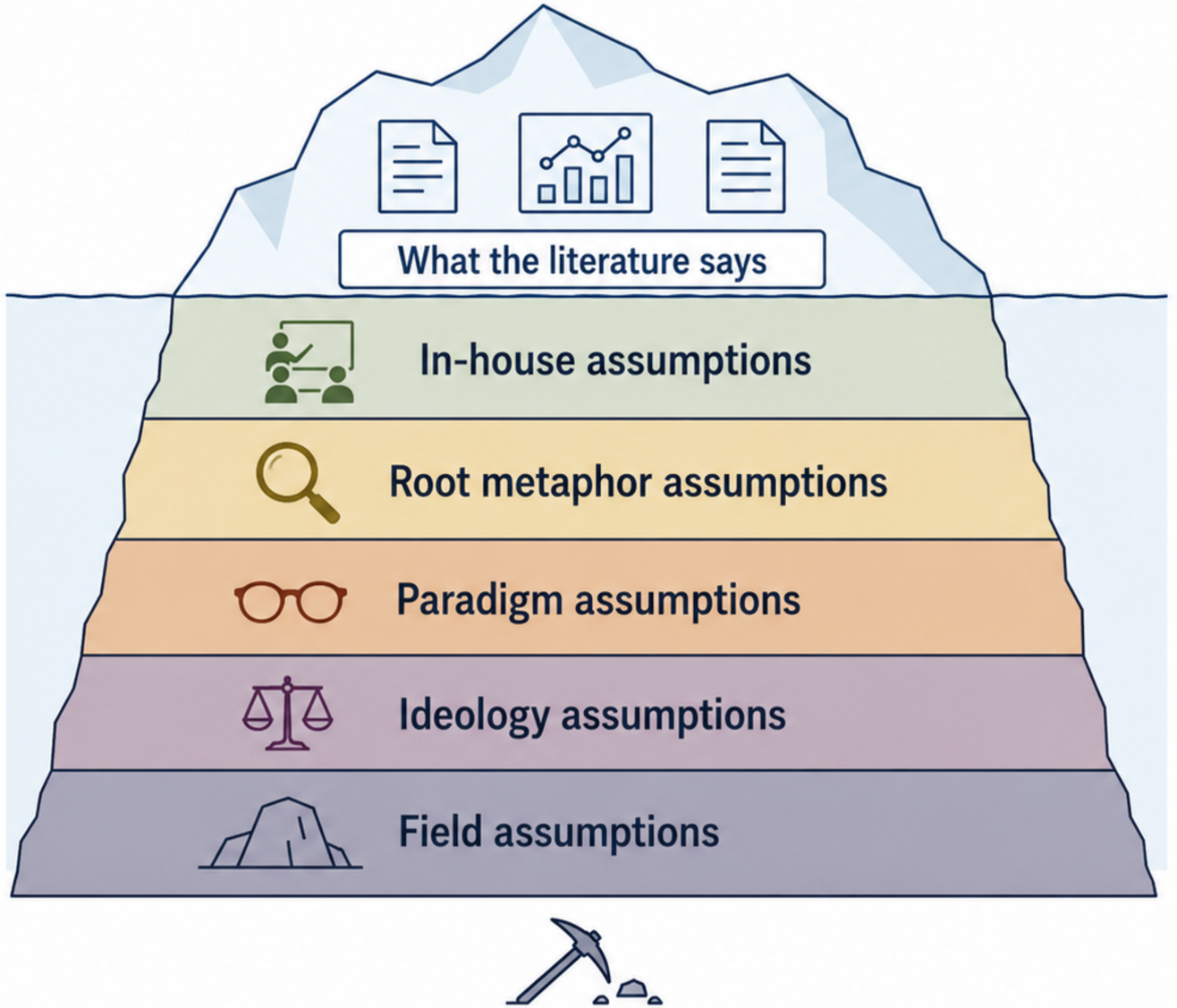
Output

New research question

What if the accepted premise is wrong?

Assumptions are layered

The deeper the assumption, the more powerful — and risky — the challenge



Our ‘school’ assumes…

This is like…

Increasing depth / scope

Reality works like…

This value matters…

Everyone assumes…

Increasing potential impact — but also difficulty and risk

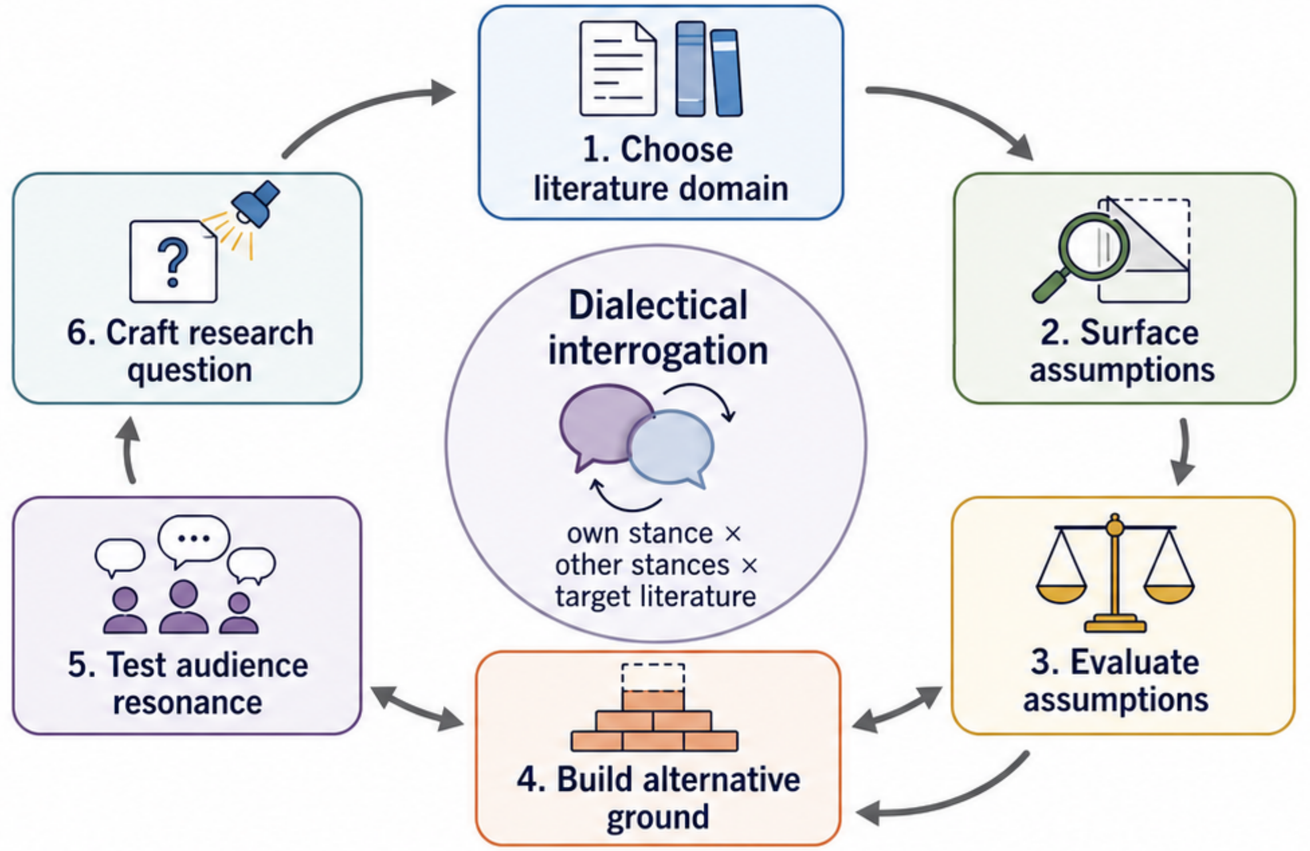
Where is the assumption worth challenging?

SE layered assumptions!?

- **In-house:** TDD research assumes tests-first is the key causal ingredient.
- **Root metaphor:** Technical debt treats poor code as financial debt.
- **Paradigm:** Controlled experiments assume SE practices can be isolated and measured.
- **Ideology:** Automation research often assumes less human effort is better.
- **Field:** SE broadly assumes software quality can be improved through better processes/tools.

The Problematization Loop

How to problematize without just criticizing



Ask colleagues: surprising but defensible?

Not a recipe

Requires judgment, creativity, and timing

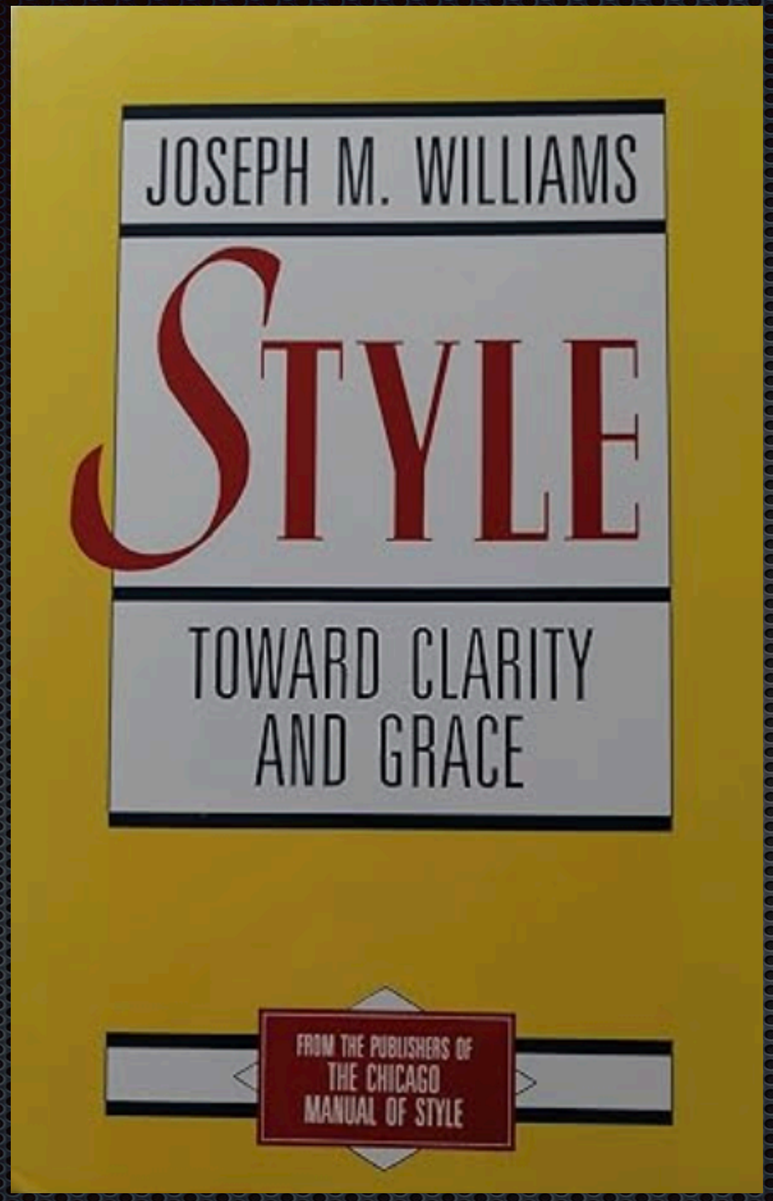
Underproblematization
Too little challenge; missed opportunities

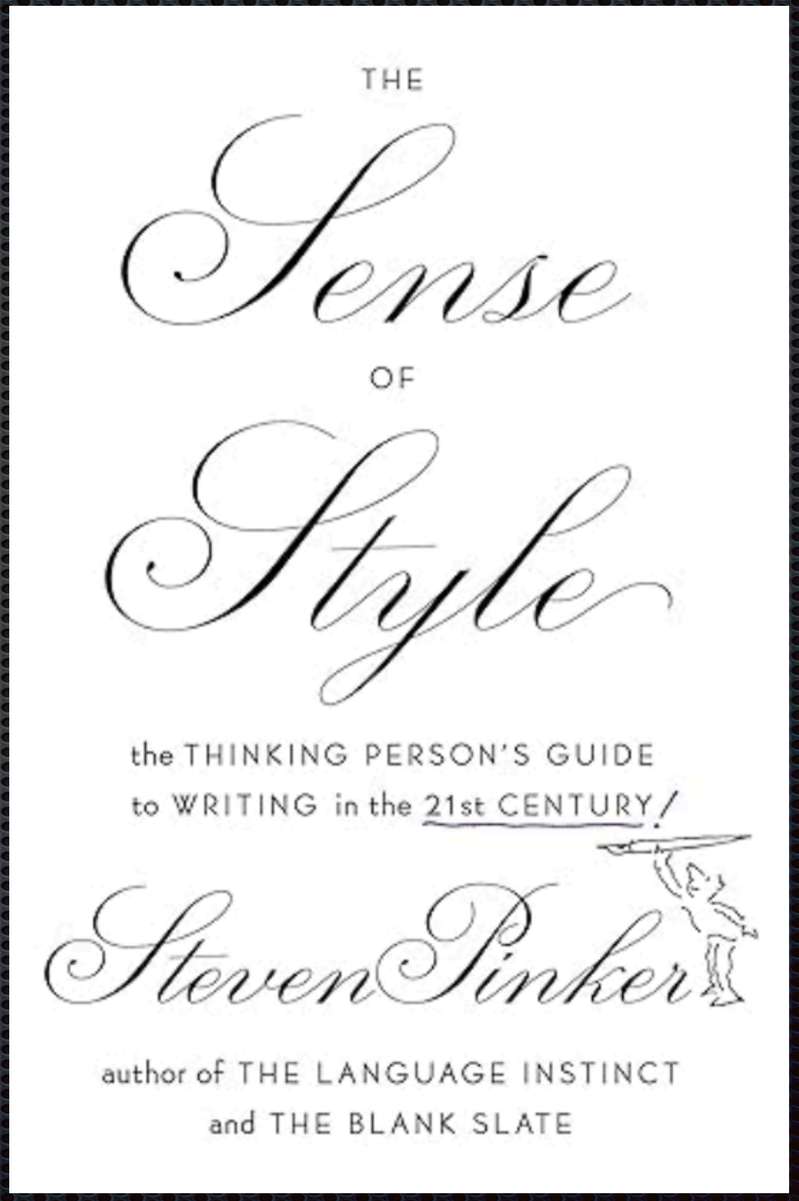
Productive challenge
Meaningful, rigorous, and constructive

Overproblematization
Excessive critique; paralysis or rejection

From Gap-spotting to Problematising RQs in SE!?

- **Gap-spotting RQs (extends): What is missing in existing research?**
  - How common is flaky testing in open-source projects?
  - Does LLM-based code review improve defect detection?
  - What factors influence adoption of CI/CD in small companies?
- **Problematizing RQs (can reshape): What if we challenge one accepted assumption?**
  - When does flaky testing reveal real system instability rather than test-suite noise?
  - How does LLM-based code review shift accountability between developer, reviewer, and tool?
  - Under what organizational conditions does CI/CD reduce rather than increase delivery control?


JOSEPH M. WILLIAMS
STYLE
TOWARD CLARITY AND GRACE
FROM THE PUBLISHERS OF THE CHICAGO MANUAL OF STYLE


THE
Sense
OF
Style
the THINKING PERSON'S GUIDE
to WRITING in the 21st CENTURY
Steven Pinker
author of THE LANGUAGE INSTINCT
and THE BLANK SLATE

Mensch & Kording's 10 rules for structuring papers (4 Principles)

- 1. Focus on a central contribution, which you communicate in the title
- 2. Write for flesh-and-blood human beings who do not know your work
  - You are expert on your work so least qualified to judge your writing from perspective of a naive/non-knowing reader
- 3. Stick to the context-content-conclusion (C-C-C) scheme
  - Whole-paper: intro sets context, results is content, discussion brings home the conclusion
  - Paragraph: 1st sentence defines context, body is content, last sent. provides conclusion
- 4. Optimize your logical flow by avoiding zig-zag and using parallelism
  - Zigzag: only central idea of paper should be touched upon multiple times
  - Parallelism: parallel messages should be communicated with parallel form

Mensch & Kording's 10 rules for structuring papers (4 Components)

- 5. Tell a complete story in the abstract
  - Context = intro to field + sub-question, specific missing gap, why important
  - Content = method/approach + executive summary of result
  - Conclusion = interprets results and answers question from context
- 6. Communicate why paper matters in the introduction
  - progressively more specific paragraphs to clarify what is lacking in the literature, followed by a paragraph summarising what is done to fill gap.
- 7. Deliver results as sequence of statements that logically connect to support central contribution
  - convince the reader that the central claim is supported by data and logic

Mensch & Kording's 10 rules for structuring papers (4th Comp.)

- 8. Discuss how the gap was filled, the limitations of the interpretation, and the relevance to the field
  - typically done by recapitulating the results, discussing the limitations, and then revealing how the central contribution may catalyze future progress.
  - first discussion paragraph is special in that it generally summarizes the important findings from the results section.Context = intro to field + sub-question, specific missing gap, why important

Mensch & Kording's 10 rules for structuring papers (2 Process)

- 9. Allocate time where it matters: Title, abstract, figures, and outlining
  - The central logic that underlies a scientific claim is paramount, outline as you go along
  - Title, abstract, and figures are viewed by far more people than the rest of the paper
- 10. Get feedback to reduce, reuse, and recycle the story
  - Use test readers to check that your story and paper works; apply checklist

Mensch & Kording's 10 rules for structuring papers (Checklist)

Rule	Sign it is violated
1: Focus on one big idea	Readers cannot give 1-sentence summary.
2: Write for naive humans	Readers do not "get" the paper.
3: Use context, content, conclusion structure	Readers ask why something matters or what it means.
4: Optimize logical flow	Readers stumble on a small section of the text.
5: Abstract: Compact summary of paper	Readers cannot give the "elevator pitch" of your work after reading it.
6: Introduction: Why the paper matters	Readers show little interest in the paper.
7: Results: Why the conclusion is justified	Readers do not agree with your conclusion.
8: Discussion: Preempt criticism, give future impact	Readers are left with unanswered criticisms and/or questions on their mind.
9: Allocate time wisely	Readers struggle to understand your central contribution despite your having worked hard.
10: Iterate the story	The paper's contribution is rejected by test readers, editors, or reviewers.

https://doi.org/10.1371/journal.pcbi.1005619.t001

The Task of the Referee

Alan Jay Smith

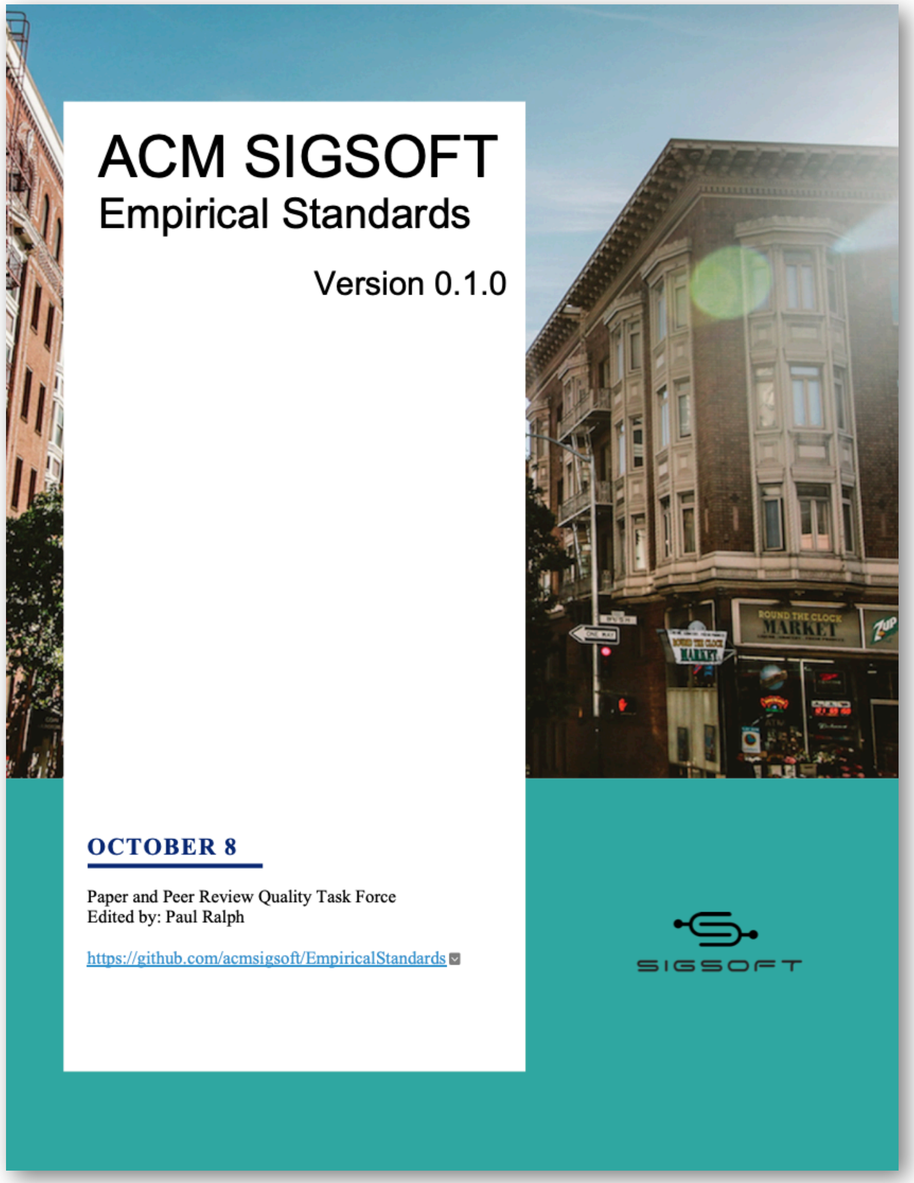
University of California at Berkeley

The referee's task

Your role as referee is to decide whether a paper makes a *sufficient contribution* to the field. The contribution can be new and interesting research results, a new and insightful synthesis of existing results, a useful survey of or tutorial on a field, or a combination of those types. To quote a referee for this article:

> Small results which are surprising and might spark new research should be published; papers which are mostly repetitions of other papers should not; papers which have good ideas badly expressed should not be published but the authors should be encouraged to rewrite them in a better, more comprehensible fashion.

Reading a paper as a referee is closer to what a teacher or professor does when grading a paper than what a scientist or engineer does when reading a published work. In the latter case, the reader presumes that the paper has been checked (refereed) and is thus correct, novel, and worthwhile. As a referee, on the other hand, you must read the paper carefully and with an open mind, checking and evaluating the material with no presumption as to its quality or accuracy. The result



https://www2.sigsoft.org/EmpiricalStandards/tools/

General standard:
https://www2.sigsoft.org/EmpiricalStandards/docs/?standard=GeneralStandard

Good Review Practices

Reviewers evaluate a manuscripts' trustworthiness, importance and clarity. The results must be, primarily, true (trustworthy) and, secondarily, important. A paper that is trustworthy can be accepted even if it is not important. A paper that is not trustworthy cannot be accepted, even if it seems important. Papers that are both trustworthy and important can have priority. Papers must be clear enough to judge their trustworthiness and importance. Reviewers should endeavor to:

- Reflect on and clearly state their own limitations and biases.
- Clarify which are necessary and which are suggested changes. Ideally, separate them.
- Identify parts of the paper that you cannot effectively judge or did not review.

Reviewing Antipatterns

- Applying empirical standards in a mechanical, inflexible, box-ticking or gotcha-like manner.
- Rejecting a study because it uses a methodology for which no specific standard is available.
- Skimming a manuscript instead of carefully reading each word and inspecting each figure and table.
- Unprofessional or vitriolic tone, ad hominem attacks, disparaging or denigrating comments.
- Allowing the authors' identities or affiliations to affect the review.
- Focusing on superficial details of paper without engaging with its main claims or results.
- Requesting additional analysis not directly related to a study's purpose or research question, leading to results poorly linked to the article's narrative.
- Using sub-reviewers when the venue does not explicitly allow it.
- Using the review to promote the reviewer's own views, theories, methods, or publications.

From ICSE 2024 PC guidelines (credit go to ICSE PC Chairs):

What fits in ICSE, and what does not?

We will consider a paper to be **in scope if it brings insights to the field of software engineering**

Papers contributing to other fields (e.g., machine learning, compilers, hardware) **without a clear link to software engineering will be considered out of scope**

In the following, we provide details and examples of what would be expected for papers belonging to different sub-areas of software engineering

Note: some areas (software evolution, in particular) may not have specific restrictions as they inherently represent software engineering topics

From ICSE 2024 PC guidlines:

Area: Artificial Intelligence and Software Engineering - *outside scope examples*

- A paper that designs a new AI algorithm with no clear link to software engineering or clear justification how it will advance the state of research and/or practice in software engineering
- A paper that applies an existing AI algorithm to non-software engineering data with no clear link to software engineering or clear justification for how it will advance the state of research and/or practice in software engineering
- A paper that provides a theoretical characterization of a property of an AI algorithm with no clear link to software engineering or clear justification for how it will advance the state of research and/or practice in software engineering
- A paper that presents a method claimed to be designed for AI-based systems without describing clearly the context within the software engineering process, task and/or tool in which the method will be used for building/maintaining/testing AI-based systems

From ICSE 2024 PC guidlines:

Review — *A Strategy*

1. Remember to refer to this slide when reviewing. Bookmark it!
2. Before reading a paper, remember the **reviewing criteria** by reviewing these slides.
3. Read the paper, and as you do, note **positive and negative aspects** for each of the five criteria.
4. Use your notes to **outline a review organized by the five criteria**, so authors can understand your judgments for each criteria.
5. **Draft your review** based on your outline.
6. Edit your review, making it as **constructive and clear as possible.** Even a very negative review should be respectful to the author(s), helping to educate them.
7. Based on your review and your assessment of the individual criteria, choose a **recommendation score**.

From ICSE 2024 PC guidlines:

Review — *Novelty*

The novelty and innovativeness of contributed solutions, problem formulations, methodologies, theories and/or evaluations, i.e., the extent to which the paper is sufficiently original with respect to state-of-the-art.

Grounded in adequate review of prior work in a respective topic, it is up to the authors to convince you that the discoveries advance our knowledge in some way, whether it sheds more light on prior work, or adds a significant new insight. Note

- A novel idea with great potential can make a very valuable paper even with only preliminary evaluation, whereas
- An incremental idea might need more support.

From ICSE 2024 PC guidlines:

Review — *Rigor*

The soundness, clarity and depth of a technical or theoretical contribution, and the level of thoroughness and completeness of an evaluation.

The paper should answer the questions it poses, and it should do so with rigor in its research methodology (including choosing an appropriate research methodology and procedures). This is an important difference between research papers and other kinds of knowledge sharing (e.g., experience reports), and the source of certainty researchers can offer. In that sense, rigor goes *beyond* checking soundness of paper. It would also be a check on whether the authors have though the problem through.

From ICSE 2024 PC guidlines:

Review — *Relevance*

The extent to which the paper's contributions beyond prior work in terms of implications for software engineering research and practice, and if needed, under which assumptions

In all generality, impact relates to advances in the practice of software engineering (including making software less costly, more maintainable, more reliable, more reusable, safer, more secure, more usable … – this is not an exhaustive list)

Note that it is the **authors' responsibility** to explain and interpret the relevance of contributions, why they matter, what their potential implications will be, and under which assumptions.

From ICSE 2024 PC guidlines:

Review — *Relevance*

Please take note of the following while evaluating relevance.

- **Take the perspective of the targeted stakeholder**: How would this advance our knowledge? How could this impact my work? Under which assumptions?
- **Do** assess technical contributions in light of **all** involved costs and risks. Weigh reported utility against required effort for setup and maintenance.
- **Assess** technical contributions not only by their evaluation results, but also by the **potential implications** of the underlying ideas. Consider pathways to deployment
- **We welcome insights** about the **practice of software engineering**.
- **Do** consider that impact can also result through **methodological contributions.**

From ICSE 2024 PC guidlines:

Review — *Verifiability and Transparency*

The extent to which the paper includes sufficient information to understand how an innovation works; to understand how data was obtained, analyzed, and interpreted; and how the paper supports independent verification or replication of the paper's claimed contributions.

This aims to check whether the described research is *recoverable*. You should be able to understand most of the key details about how the authors conducted their work, how they made their invention possible, or how the research findings were inferred from the collected evidence. This is key for replication and meta-analysis of studies underpinned by the positivist or post-positivist approaches. For interpretivist works, it is also key for evaluating qualitative work. Focus your critiques on omissions of research process or innovation details that would significantly alter your judgement of the paper's validity, or the *credibility* of results for research that uses qualitative methods.

From ICSE 2024 PC guidlines:

Review — *Presentation*

The extent to which the paper's quality of writing meets the high standards of ICSE, including clear descriptions, as well as adequate use of the English language, absence of major ambiguity, clearly readable figures and tables, and adherence to the formatting instructions provided below.

Papers also need to be clear and concise, and comprehensible to diverse audiences.

We recognize that not all authors are fluent English writers. But if the language issues make the paper not comprehensible, it is not yet ready for publication.

Reviewing research papers: Do's and Don'ts

These patterns come from two sources. The main one is a comparison of the strongest and weakest reviewers at a recent empirical software engineering conference, judged by everything each reviewer wrote. The other is a survey of software engineering authors and reviewers about what makes reviews good and bad (Prechelt, Graziotin and Méndez Fernández, 2018). **[E]** = essential: this is what separated the top reviews from the bottom ones. **[C]** = contributes, but matters less. Both lists are in rough priority order.

TL;DR

Do

1. **[E]** Test the paper's core claims against its evidence
2. **[E]** Judge the work by the standards of its own research method
3. **[E]** Anchor every point in the paper (section, line, table, quote)
4. **[E]** Verify things yourself: numbers, tables, citations, the replication package
5. **[E]** Separate verdict-changing issues from minor ones
6. **[E]** Make every criticism actionable and realistically scoped
7. **[E]** Make your score consistent with your text
8. **[E]** Be fair: credit strengths and critique the work, not the authors
9. **[E]** Keep the same care across all your reviews, including the positive ones
10. **[C]** Check for overclaiming
11. **[C]** Structure by the venue's review criteria, with a strengths/weaknesses summary
12. **[C]** Open with an accurate summary of what the paper does and claims
13. **[C]** Point to concrete literature and standards, with full references
14. **[C]** Be honest about your expertise and any uncertainty
15. **[C]** Check the venue's specific requirements in its call for papers
16. **[C]** Collect typos and presentation fixes in a separate minor list

Don’t

1. **[E]** Reject on opinion or taste without evidence from the paper
2. **[E]** Write generic or template text that could fit any paper
3. **[E]** Submit LLM-generated text you haven’t engaged with
4. **[E]** Give a strong verdict outside your competence
5. **[E]** Write reviews too short to guide the authors
6. **[E]** Be dismissive, sarcastic or condescending
7. **[E]** Hide your substantive critique in the confidential comments to the programme committee
8. **[C]** Make accusations without showing the evidence
9. **[C]** Reject mainly on scope or fit without an argument
10. **[C]** Push your own work
11. **[C]** Treat paper structure and outline conventions as a main flaw
12. **[C]** Bury the main issues under clarification questions and nitpicks
13. **[C]** Use shaky methodological reasoning yourself
14. **[C]** Submit unfinished reviews